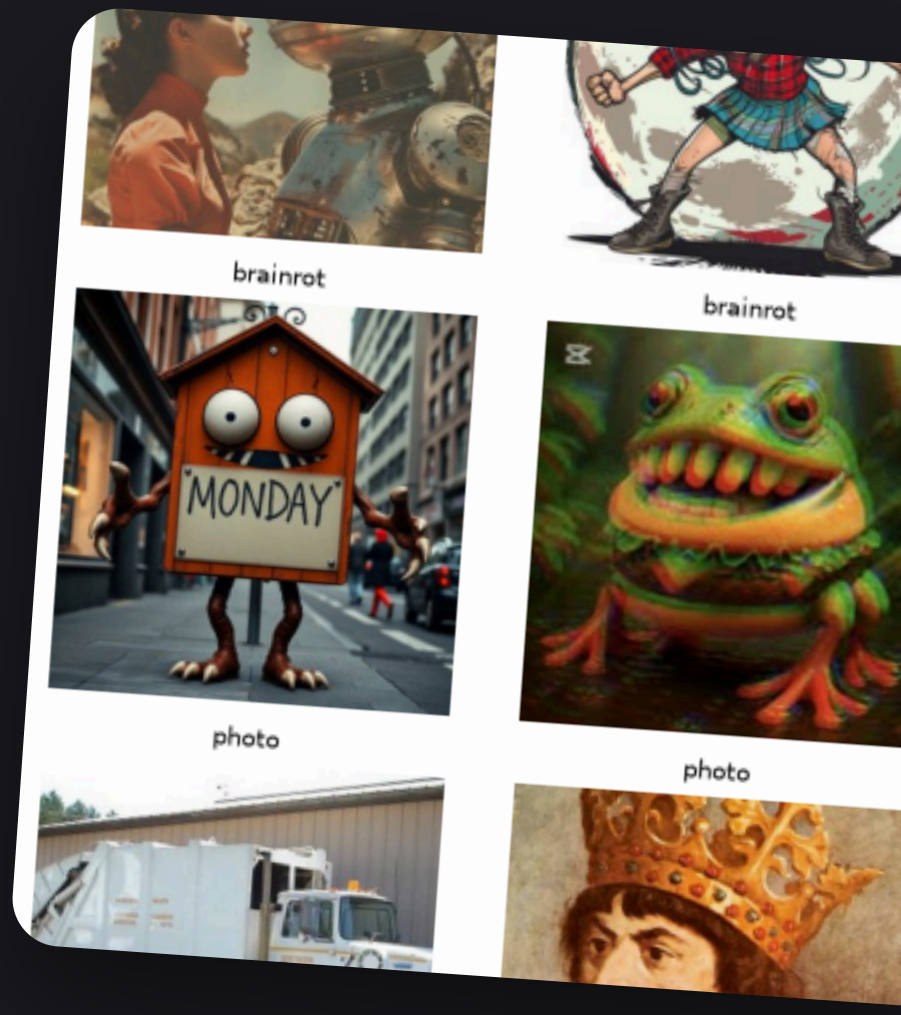
MACHINE LEARNING · NAUKMA · MINI-PROJECT

Brainrot Detector

Is this image an Italian brainrot character? A binary image classifier — and an honest look at what it actually learnt.



Team

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Code, notebook and model: ML_Brainrot repository

WORK DONE

An end-to-end study: from raw downloads to a working demo app

01
Data pipeline
Download 3 public sources (~700 MB), standardise every image identically, hash-dedupe, split stratified by source.

02
Baselines
Majority class (floor) and HSV colour histogram + logistic regression to measure "style only".

03
Transfer learning
Frozen MobileNetV2 embeddings + logreg; then MobileNetV2 fine-tuned in Keras (head → top 30 layers) with augmentation.

04
Validation
5-fold stratified CV for sklearn models; early stopping + threshold on val for Keras; test set used once. Bootstrap 95% CIs.

05
Error analysis
Errors per source, confident mistakes, border-removal test, occlusion maps, hard-negative learning curve.

06
Demo app
Streamlit app: upload an image, get brainrot probability with adjustable threshold and test metrics.

9 982
images from 3 public sources, after deduplication

4
models compared, from majority class to fine-tuned CNN

5
error-analysis checks of what the model really learnt

~15 min
full notebook on a laptop CPU · seed 42, pinned versions

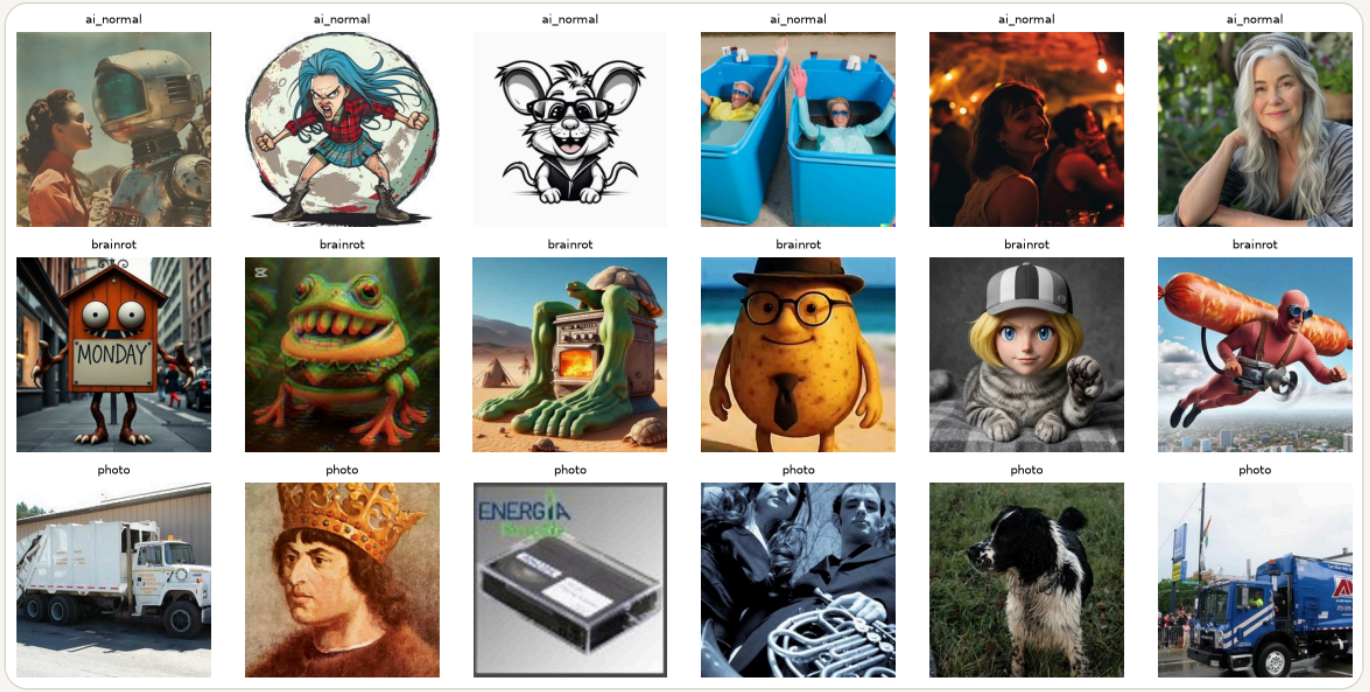
THE QUESTION

Every brainrot image is AI-generated — so we added ordinary AI images as hard negatives

If the "not brainrot" class were only real photos, a model could score well by simply detecting AI images. To test what it really learns, class 0 mixes photos with Midjourney / DALL·E / SD images, and every result is reported **per source**.

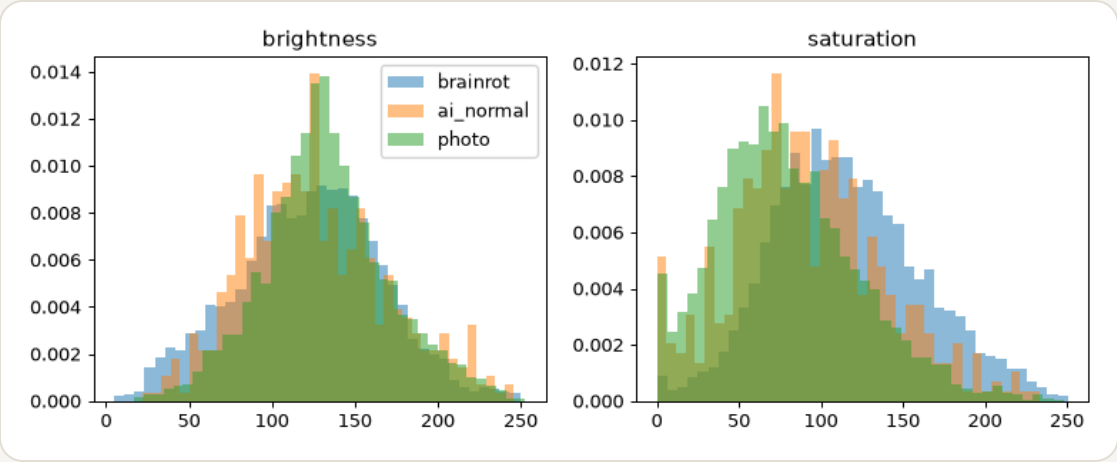
SOURCE	LABEL	IMAGES
brainrot	1	4 982
photo	0	4 500
ai_normal	0	500
total · split 70 / 15 / 15		**9 982**

Kaggle Italian Brainrot dataset · HF midjourney-dalle-sd · Imagenette. All images → RGB, centre crop, 256², JPEG q90; 18 duplicates removed before splitting.



DATA EXPLORATION

Patterns a model could learn, and the shortcuts we removed



Per-image mean HSV brightness and saturation (0–255), density per source.

AVERAGE PER SOURCE	BRIGHTNESS	SATURATION
brainrot	122.5	115.9
ai_normal	126.5	91.9
photo	132.5	79.4

Signals in the data

Colour saturation REAL SIGNAL

Brainrot images are clearly more colourful (116 vs 79 for photos). Our colour-histogram baseline tests how far this alone goes: ROC-AUC 0.77.

Brightness WEAK

Distributions overlap almost completely, so brightness says little about the class.

"AI-generated look" RISKY

Every brainrot image is AI art, so "looks generated" would separate it from photos. We added ordinary AI images so this cue alone is not enough.

Shortcuts we removed or tested

- **Size, aspect ratio, compression:** every image cropped to a 256² square and saved as JPEG q90
- **Near-duplicates:** 18 removed by image hash before splitting
- **TikTok watermarks and captions at the edges:** checked later with the border-removal test

VALIDATION SCHEME & METRICS

No leakage, one look at the test set, and metrics that show where the model fails

How the data is split



- **Deduplicate first.** 18 near-duplicates (same 64-bit image hash) removed *before* splitting, so no picture is in two splits.
- **Stratified by source.** Images are independent (no users, no time order), so a random split is valid; stratifying by source keeps the 10 % share of hard negatives in every split and fold.
- **scikit-learn models:** 5-fold stratified CV on train + val (8 484 images). Scaling sits inside a `Pipeline`, so it is fitted within each fold only.
- **Keras model:** trained on train; early stopping and the decision threshold (0.35, max. balanced accuracy) chosen on val.
- **Test set used once**, for the final comparison.

Which metrics, and why

METRIC	WHY WE USE IT
ROC-AUC	Threshold-free ranking quality; main number for comparing models.
PR-AUC	Focuses on the positive (brainrot) class.
F1, balanced acc.	Quality at the chosen threshold, i.e. what the app actually does. Classes are ~50/50, but balanced accuracy stays honest if that changes.
Error rate per source	The key diagnostic. Class 0 is 90 % easy photos and 10 % hard AI images, so overall scores can hide failures on the hard ones.
95% confidence intervals	Bootstrap CI for ROC-AUC; Wilson CI for the small 75-image hard-negative group.

RESULTS · HELD-OUT TEST SET (1 498 IMAGES)

Each model beats the last; fine-tuned MobileNetV2 reaches ROC-AUC 0.986

MODEL	ROC-AUC	BAL. ACC.	F1	BRAINROT MISSED	PHOTO FALSE ALARM	AI_NORMAL FALSE ALARM
Majority class	0.500	0.500	0.000	100 %	0 %	0 %
Colour histogram + logreg	0.773	0.717	0.725	25.3 %	28.9 %	53.3 %
MobileNetV2 embeddings + logreg	0.972	0.931	0.931	7.2 %	3.0 %	38.7 %
Fine-tuned MobileNetV2	**0.986**	**0.951**	**0.951**	**4.7 %**	**1.5 %**	**38.7 %**

0.77

ROC-AUC from **colour alone** — brainrot has a recognisable visual style.

1.5 %

of everyday photos wrongly flagged — a usable "brainrot vs photo" filter.

38.7 %

of ordinary AI images wrongly flagged — the weak spot (next slide).

ROC-AUC 95% CI for fine-tuned model: 0.981–0.991 · threshold 0.35 chosen on validation · 5-fold CV agrees (embeddings: 0.978 ± 0.002)

FINE-TUNING MOBILENETV2

Fine-tuning helped on brainrot and photos, not on AI images

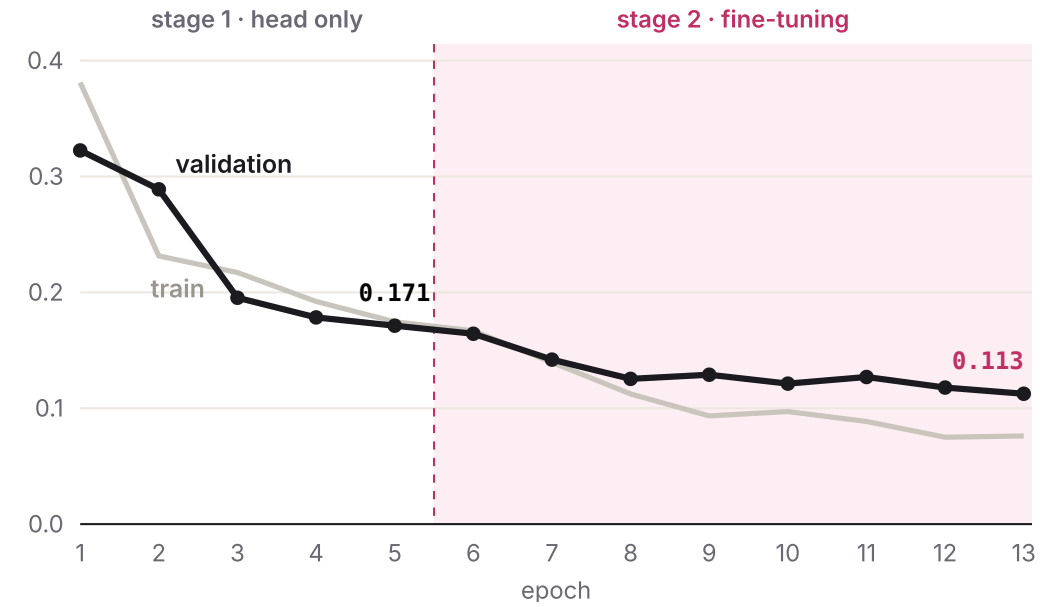
What we did

1. **Stage 1: head only.** Backbone frozen; dropout 0.2 + sigmoid layer trained with lr 1e-3 for 5 epochs.
2. **Stage 2: top 30 layers unfrozen.** 100× smaller lr (1e-5), so ImageNet features are adjusted, not destroyed.
3. **Safeguards.** BatchNorm kept frozen, augmentation (flip, rotate, zoom, contrast), early stopping on val loss.

Impact on the test set

	FROZEN FEATURES + LOGREG	FINE-TUNED	CHANGE
ROC-AUC	0.972	0.986	+0.014
Balanced accuracy	0.931	0.951	+0.020
brainrot missed	7.2 %	4.7 %	−35 %
photo false alarm	3.0 %	1.5 %	−50 %
ai_normal false alarm	38.7 %	38.7 %	none

Loss during training



In stage 2, val loss fell another **34 %** and val accuracy rose from 93.5 % to **95.9 %**. Early stopping never triggered: val loss was still falling at the last epoch, so longer training might help a little.

KEY FINDING

The model partly learnt "AI-generated cartoon character ⇒ brainrot"

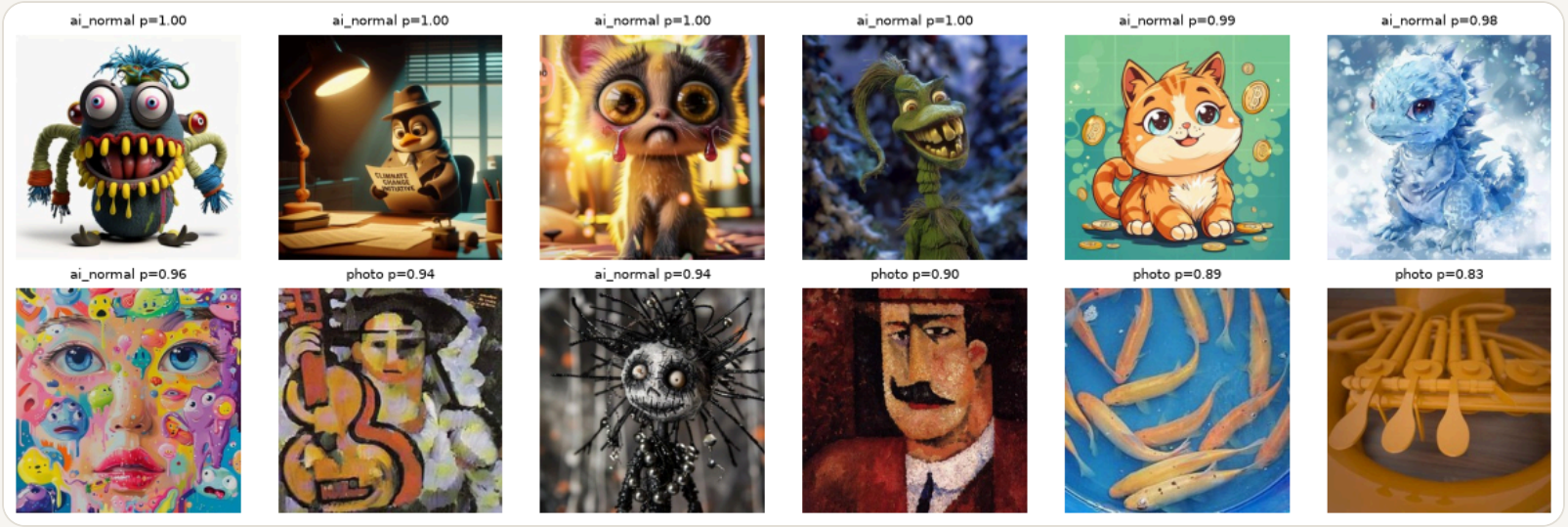
29 / 75

ordinary AI test images flagged as brainrot

`38.7 % · Wilson 95% CI 28–50 %`

The false alarms are mostly cute or creepy creatures with big eyes and saturated colours — visually close to the meme genre, but not brainrot.

A few painterly photos (cubist portraits, a fish tank) are also caught by the "AI style" cue.

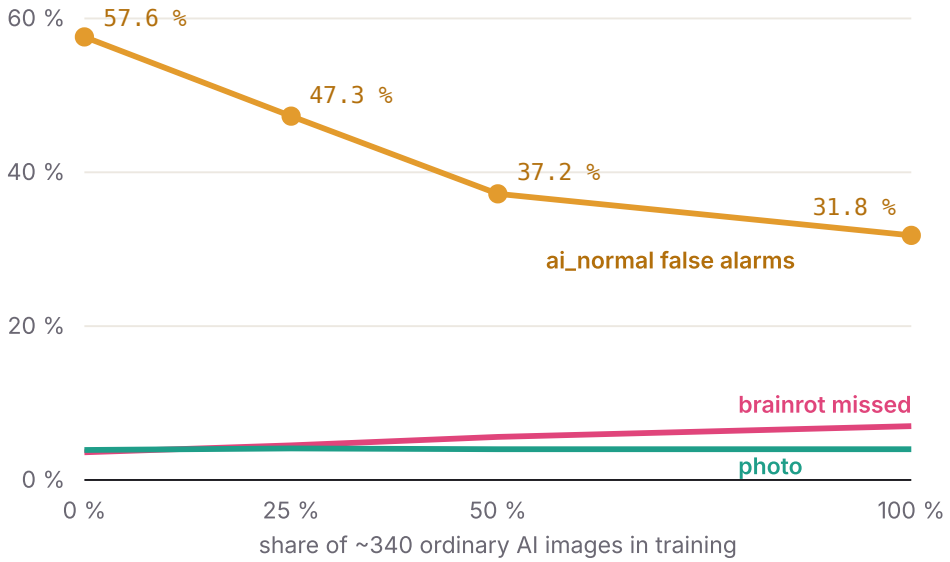


Most confident false positives on the test set (label · predicted brainrot probability).

DIAGNOSTICS

Not a watermark shortcut — and more hard negatives keep helping

Hard-negative learning curve · embeddings model, 5-fold CV



False alarms on AI images fall from 58 % to 32 % and **have not levelled off** — at a small cost in brainrot recall.

Border-removal test · keep only central 60 %

SOURCE	FULL IMAGE	CENTRE ONLY
brainrot	4.7 %	9.1 %
photo	1.5 %	3.4 %
ai_normal	38.7 %	32.0 %

Cropping hurts brainrot and photos about equally → **no sign** the model relies on TikTok watermarks or captions.

Occlusion maps

Important regions sit on the character (faces, eyes, hybrid body parts), not on the background — qualitative only.

CONCLUSIONS

Usable as a "brainrot vs photos" filter — not yet as "brainrot vs any AI picture"

Supported

- Reliable separation from photos: ROC-AUC 0.986, 1.5 % false alarms
- Most gain comes from pretrained ImageNet features
- No watermark / caption shortcut

Not supported

- That the model understands "brainrot" as a concept
- 39 % of ordinary AI images flagged

Limitations

- Only 75 hard negatives in test
- One meme wave, one wiki; 10 photo classes
- Keras model trained with one seed

Next steps

- **Collect more hard negatives** — the curve says this is the biggest lever
- Add other meme styles and non-ImageNet photos
- Repeat fine-tuning over several seeds

Demo

`streamlit run app.py` — upload an image, get the brainrot probability.